Key Points / Highlights

Manuscript: Deep Learning Architectures for Peptide-MHC Class I Binding Prediction

Journal: *Briefings in Bioinformatics*, Oxford University Press

---

1. Six deep learning architectures were systematically benchmarked for HLA-A*02:01 peptide binding prediction under controlled conditions; the Deep Feed-Forward Network with batch normalization achieved the highest performance (91.9% accuracy, ROC AUC 0.947).
2. Label quality, not model architecture, defines the performance ceiling: deterministic PSSM labels inflate accuracy by ~3 percentage points relative to machine-learning-derived labels, with the weak binder class as the primary locus of variation — a finding with significant implications for benchmarking studies.
3. The best model achieves 93.9% sensitivity against 49 experimentally validated IEDB T-cell epitopes, and with a sequence-diversity homopolymer filter, 100% specificity — formalizing the binding-versus-immunogenicity distinction.
4. Cancer hotspot mutation scanning of p53 and KRAS identifies actionable neoepitope candidates: p53 R248W and KRAS G12V create novel epitopes; KRAS G12C enhances an existing epitope, suggesting combination immunotherapy strategies.
5. All code, trained models, feature matrices, and prediction results are provided as an open-source reproducible pipeline with a complete data package.